ZCAS University

MASTER OF SCIENCE IN INFORMATION TECHNOLOGY

CCS6611 - COMMUNICATIONS NETWORKS

FINAL EXAMINATION

2ND JUNE 2025

14:00 – 17:00 HOURS

TIME ALLOWED: READING AND WRITING TIME– THREE HOURS AND FIVE MINUTES

INSTRUCTIONS:

1. Section A: **Question One** in Section A is **compulsory.**
2. Sections B: Answer **TWO** Questions from this section.
3. This examination paper carries a total of **100 marks.**
4. Do Not Open the paper Until told to do so by the Invigilator

SECTION A – COMPULSORY

QUESTION ONE

(a) Study carefully the undirected graph in Figure 1 below reflecting a network structure with nodes representative of switches, and edges representing interlinks amongst them.

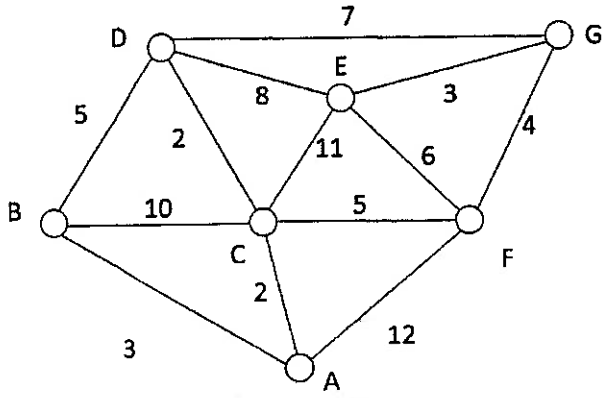


Fig. 1 Undirected graph

i. State the conditions for a Euler path and Euler circuit. **[4 Marks]**

ii. Compute the adjacency matrix and incident matrix of the network structure. **[6 Marks]**

iii. Use the Dijkstra algorithm to find shortest path from node A to G. **[5 Marks]**

iv. Based on your answer in (iii), explain why the Dijkstra algorithm is considered to be a greedy algorithm. **[5 Marks]**

v. Explain how OSPF calculates the cost of a given link (Ethernet, FastEthernet, & GigabitEthernet) using a base bandwidth of 10^8. **[5 Marks]**

(b) Consider the network of switches below in Figure 2 where STP is to be implemented and answer the questions that follow:

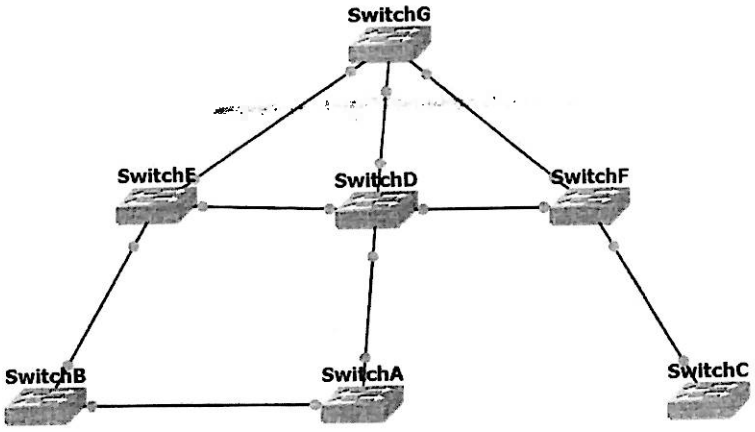


Figure 2. Switched network

i. Determine and state the number of cycles in the network by drawing a graph depicting well-labeled cycles. **[5 Marks]**

ii. Name the switch, which will be chosen as the root bridge and explain why. **[5 Marks]**

iii. Given the switch labels denote their IDs, use the Spanning Tree Protocol to redraw and label all the ports as RP, DP, and BP. **[5 Marks]**

iv. Derive and draw the resultant spanning tree. **[5 Marks]**

v. In tabular form, explain the difference between IEEE802.1d, IEEE802.1w, and IEEE802.1s. **[5Marks]**

Total: (50 Marks)

SECTION B: Answer ANY TWO Questions

QUESTION TWO

Given an IP address range 100.20.18.0/24, subnet this range with 7 usable host IPs in each subnet, and complete the table below for the first 5 subnets:

	Network ID	**Subnet Mask**	**CIDR Netmask**	**Broadcast ID**	**Wildcard Mask**
1st Network					
2nd Network					
3rd Network					
4th Network					
5th Network					

(25 Marks)

QUESTION THREE

Consider the network diagram below in Figure 3 depicting the network of a SME. It comprises of two LANs (Branch LAN & HQ LAN). The HQ LAN is the corporate network segment whereas the Branch LAN is the network of the branch.

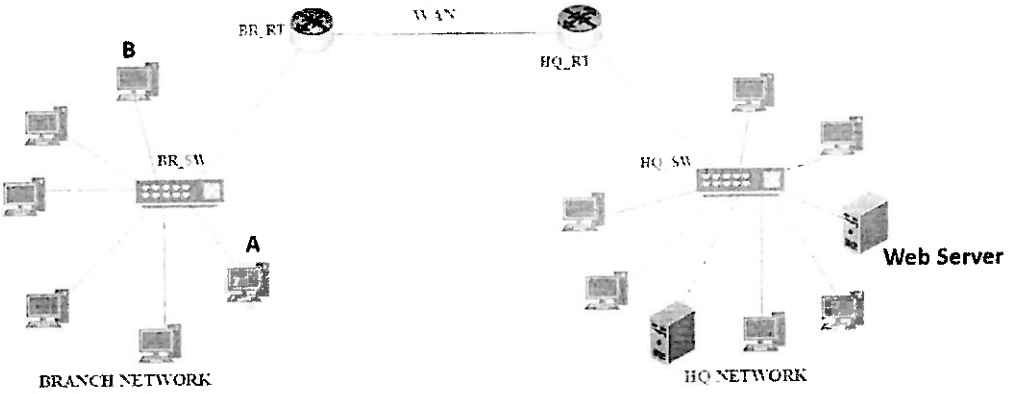


Figure 3. SME network

(a) Given that PC A in the Branch LAN wants to communicate with PC B within the same subnet, explain how the PC A figures out in which network PC B is located and what media access mechanism (protocol) it will use to communicate with it. **[10 Marks]**

(b) Given that PC A now wants to communicate with the Web Server at the HQ, explain how the PC A figures out in which network the Web Server is located and how it will proceed to communicate with it. **[10 Marks]**

(c) State and explain which type of subnet mask addressing you would use between BR_RT and HQ_RT. **[5 Marks]**

(25 Marks)

QUESTION FOUR

WiFi technology plays a crucial role in modern networking, providing the backbone for a wide range of wireless communications. It has become an essential component in both personal and professional environments. Carefully, consider the below and answer the questions that follow regarding WiFi technologies.

Table 2: WiFi Technology Characteristics.

WiFi Version	IEEE Standard	Frequency	Max Speed	Security Features
WiFi 1				
WiFi 4				
WiFi 5				
WiFi 6				
WiFi 7				

i. Fill in and complete Table 1 depicting the various characteristics of WiFi technologies. **[10 Marks]**

ii. With the help of RTS/CTS, explain the CSMA/CA media access mechanism. **[5 Marks]**

iii. Explain how WiFi (IEEE 802.11) operates across the OSI model layers, focusing on its key protocols and functions at each relevant layer. **[5 Marks]**

iv. Discuss 5 environmental factors that affect the performance of Wi-Fi networks. **[5 Marks]**

(25 Marks)

END OF EXAMINATION